

SH-205 Intelligent Shipment Piggybacking

Technical Architecture & Live Demonstration Reference Guide

Track: Autonomous Supply Chain & Logistics	Target Region: Telangana Linehaul Grid (NH 163, NH 65, NH 44)	Status: Production Ready Hackathon Pitch
Engine: Google OR-Tools CVRPTW	Dataset Scale: 200 Shipments · 32 Hubs · 75 Fleet Trucks	Avg Solution Time: < 28.5 ms

1. Executive Summary & The SH-205 Problem

Modern surface freight logistics is plagued by an expensive, fragile operational failure: **stranded, misrouted, and delayed shipments**. In intermediate transfer corridors, cargo is routinely offloaded at the wrong transit facility (e.g., misrouted to Nalgonda H07 instead of direct transit along the NH-163 corridor to Warangal H02).

Failure Mode (The Status Quo)	Autonomous Recovery (SH-205 Innovation)
• Dedicated Spot-Hire Dispatch: Dispatchers hire emergency hotshot trucks costing █35–█45/km. • Empty Deadhead Waste: Recovery trucks travel 100% empty to reach the stranded hub. • SLA Breaches: Manual rebooking takes 4–8 hours, triggering severe commercial penalties. • Cost Impact: █8,500–█12,500 per recovery incident with high carbon footprint.	• Dynamic Piggybacking: Identifies active linehaul trucks with unutilized spare capacity. • Zero Empty Deadheads: Injects stranded cargo into pre-scheduled routes passing nearby. • Sub-Second Autonomous Convergence: CVRPTW solver calculates optimal plan in < 30 ms. • Economic ROI: █6,500 net savings (76.5% reduction) and 250 kg CO█ avoided per recovery.

2. System Architecture & Tech Stack

The SH-205 platform uses a modern, decoupled microservices architecture designed for extreme responsiveness, strict geospatial precision, and transparent multi-stage visual lifecycle management:

Component	Technologies Used	Architectural Responsibility
Frontend UI & Map Engine	• React 19 / Vite • Tailwind CSS 3 • Leaflet.js (^1.9.4) • react-leaflet (^5.0.0)	Autonomous Control Center with progressive 3-stage visual state machine. Uses public OpenStreetMap vector tiles, custom SVG markers, dynamic polyline rendering, and MapController viewport invalidation.
Backend API & WebSockets	• Python 3.13 • FastAPI (ASGI) • Uvicorn Server • WebSocket Channels	RESTful endpoints handling disruption injection (/api/shipments/disrupt), candidate matching (/api/recovery/match), optimization (/api/recovery/optimize), and broadcast event synchronization.
AI Optimization Engine	• Google OR-Tools • CVRPTW Constraint Solver • NumPy & SciPy	Executes multi-objective mathematical optimization evaluating capacity constraints, time windows, detour distances, and fuel-cost trade-offs.
Database & Geospatial Layer	• PostgreSQL 16 • PostGIS Extension • Supabase DB / REST	Persistent storage of 200 shipments, 75 linehaul trucks, and 32 regional transit hubs. Enforces geospatial proximity indexing and atomic state persistence upon plan acceptance.

3. The AI Optimization Engine (Algorithmic Logic for Judges)

When explaining the AI solver to the jury, present the system as a **three-step deterministic decision pipeline** backed by Google OR-Tools Capacitated Vehicle Routing with Time Windows (CVRPTW):

Step A: Hard Capacity Feasibility Filtering

The engine first queries the active fleet database to enforce the fundamental physical payload constraint: `truck.Available_Capacity_kg >= shipment.Weight_kg`. For instance, stranded shipment **SH004** weighs **4,500 kg (4.5 Tons)**. The solver scans candidates and isolates vehicle **TRK-004 (V004)**, which has a 14.0T rating, currently carries 6.0T base freight, and holds **8.0 Tons (8,000 kg) of verified spare capacity**. Payload utilization expands safely from 42.8% to 75.0% without violating axle weight regulations.

Step B: Geographic Corridor Trajectory Alignment

Instead of dispatching trucks from distant depots, the algorithm evaluates vehicles whose pre-scheduled route corridors already traverse the stranded consignment's vicinity. In our primary scenario, TRK-004 is travelling along the route [H06 Mahbubnagar → H01 Hyderabad → H07 Nalgonda → H02 Warangal]. Because Nalgonda (H07) is already a scheduled waypoint on TRK-004's transit manifest, the detour addition is **0.0 km**, achieving a 100% trajectory match.

Step C: Multi-Objective Recovery Economics Formulation

The optimization objective function minimizes total operational penalty across three weighted dimensions:
$$\min Z = w_{\text{cost}} \cdot \Delta C + w_{\text{time}} \cdot \Delta T + w_{\text{carbon}} \cdot \Delta E$$
 The economic parameters are derived directly from the cost configuration matrix:

Metric Dimension	Spot-Hire Benchmark	SH-205 Piggyback Engine	Net Savings Benefit
Cost of Recovery (■)	■8,500 (Dedicated Truck @ ■35/km + handling)	■2,000 (Incremental handling + fuel surcharge)	■6,500 Saved (76.5% Direct Savings)
Added Road Mileage	186.8 km (Empty round-trip deadhead)	0.0 km – 16.8 km (Corridor intercept)	170+ km Deadheading Eliminated
CO■ Carbon Emissions	310 kg CO■ (Dedicated diesel burn)	60 kg CO■ (Incremental load resistance)	250 kg CO■ Avoided
SLA Delivery Compliance	Risk of 4–8 hr delay for vehicle dispatch	On-schedule pickup during scheduled truck stop	Delivered 2.0 hrs ahead of deadline

4. The Dataset (Telangana Logistics Network Scale)

Our system is not a synthetic toy model—it operates on a comprehensive enterprise dataset mirroring real-world inter-district logistics operations across Telangana and surrounding corridors:

32 Regional Hubs	75 Linehaul Trucks	Cost Matrix & Rates
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• Primary Hubs: Hyderabad (H01), Warangal (H02), Nizamabad (H03), Karimnagar (H04), Khammam (H05). • Intermediate Transfer Hubs: Nalgonda (H07), Siddipet (H10), Suryapet (H14), Bhongir (H16), Sangareddy (H18). • Geospatial Precision: Exact PostGIS coordinates snapped to NH 163, NH 65, NH 44, and SH 24.	• Heavy Duty: 14.0T Multi-Axle Trucks (e.g. TRK-004 with 8.0T spare). • Medium Duty: 7.5T & 10.0T Medium Commercial Vehicles (MCV). • Feeder Fleet: 3.5T & 5.0T Light Commercial Vehicles (LCV). • Telemetry: Real-time GPS location, driver ID, route manifest, and cargo load tracking.	• Operating Cost: █40–█50 per km linehaul rate. • Driver Base Pay: █800–█1,200 per transit shift. • Hub Handling Fee: █400–█500 per transshipment. • Spot Hire Baseline: █35.0/km dedicated deadhead rate + █2,000 emergency surcharge.
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5. The Live Demo Flow (Step-by-Step Script for the Pitch)

Follow this exact 4-step sequence during the live hackathon presentation to demonstrate technical mastery:

Step & UI Location	Action to Take	Visual Screen State & What to Tell Judges
Step 1: Clean Startup /dashboard	Show the initial executive Control Center.	State 1: Zero Initial Anomalies. Point out that the grid begins clean with 0 stranded consignments. The map renders the standard blue intended corridor connecting Hyderabad (H01) and Warangal (H02).
Step 2: Disruption Injection /disruption-engine	Navigate to Disruption Engine. Select SH004 (Tata Motors, 4.5T) and target hub Nalgonda (H07) . Click 'Inject Disruption'.	State 2: Transit Anomaly Generated. Spatial constraint enforces intermediate transfer (~50km off destination). Database persistently executes UPDATE shipments SET shipment_status='Misplaced'. Auto-navigates to Control Tower: map clears intended line, showing █ pulsing stranded marker at H07 and top 3 nearest candidate trucks with amber proximity badges.
Step 3: AI Recovery Solver Control Tower Header	Click '█ Run AI Recovery'.	State 3: Algorithmic Convergence. Google OR-Tools CVRPTW solver executes in < 30 ms. Identifies TRK-004 with 8.0T spare capacity. Map morphs dynamically to display the three diagnostic polylines: • █ Red Polyline: 93.4 km deviation path (H01 → H07) • █ Blue Dashed Line: Original scheduled route (H01 → H02) • █ Green Polyline: Optimized piggyback recovery (H07 → H02)
Step 4: Acceptance & Persistence Action Bar	Click 'Accept Changes'.	State 4: Permanent State Locking. The database is atomically updated (status='Recovered', is_recovery_accepted=true). The map persistently retains all 3 polylines (Red, Blue, Green) for continuous visual context. KPI card increments Total Cost Saved to +█6,500 and avoided carbon by 250 kg.

Summary for the Judges: SH-205 transforms logistics disruption from an expensive manual crisis into an automated, zero-deadhead optimization opportunity. By utilizing real linehaul telemetry, PostgreSQL/PostGIS persistence, and Google OR-Tools, the system achieves **76.5% cost reduction**, eliminates secondary vehicles, and preserves 100% SLA integrity.